Labor Market Sorting over the Business Cycle

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Introduction

- In this project, I study how the composition of unemployed workers varies over the business cycle.
- Using CPS and NLSY, I argue that workers with lower expected match surplus flow more into unemployment in recession periods
  - Using CPS microdata, I show workers with high wage and short tenure are more likely to separate from their jobs during the recession compared to normal times
  - Then, I use NLSY79 data to suggest that young workers with separation experience are more vulnerable to cyclical variations in separation probabilities than workers without such experiences
  - There is no systematic differences in job-finding rates across wage/tenure groups over the business cycle.

Introduction - continued

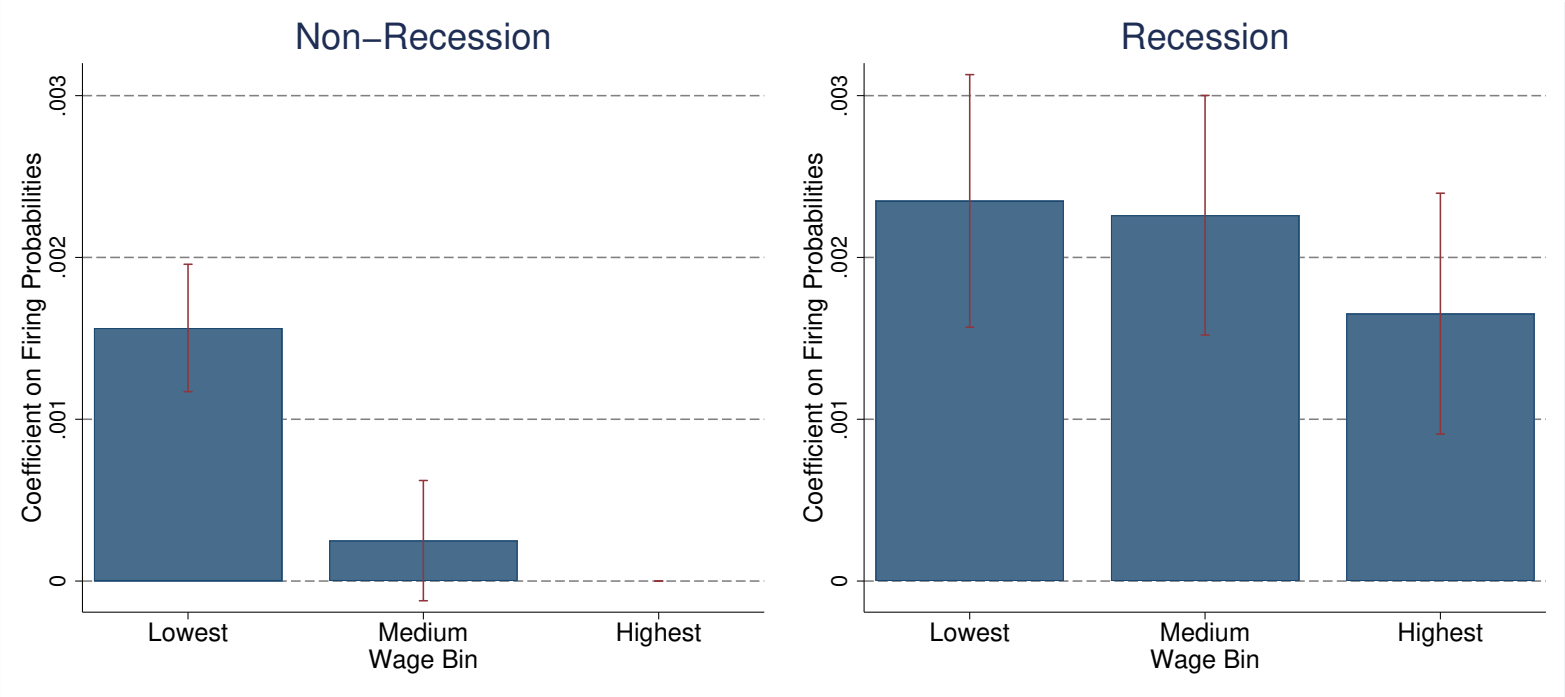
- I then construct a parsimonious Diamond-Mortensen-Pissarides model with worker types to match the data patterns
- The model is used to quantify the impact of labor market sorting on job creation over the business cycle
  - Introducing composition change leads to hump-shaped responses in job finding rates and greater fluctuations in unemployment rates
- I plan to use model estimates the replicate the cyclical cost of job-losing.

Data and Definition

- Data: CPS ASEC and Job Tenure Supplement
- Dependent variable:
  - 1 if employed in this period and becomes job loser in the next month
  - 0 if employed in this period and employed in the next month
- Key variables:
  - Wage bin dummy:
    - 3× wage bins, where workers are assigned by their rank in each year and occupation cell
    - Excluded top and bottom 5 percent in each year and occupation cell
  - Job tenure bin dummy :
    - 3× job tenure bins, where workers are assigned by their rank in each occupation cell
    - Excluded workers with job tenure < 0.1 or > 20
- Controls:
  - Age, Age squared, Race, Occupation, Industry, Gender, Education

Summary Statistics

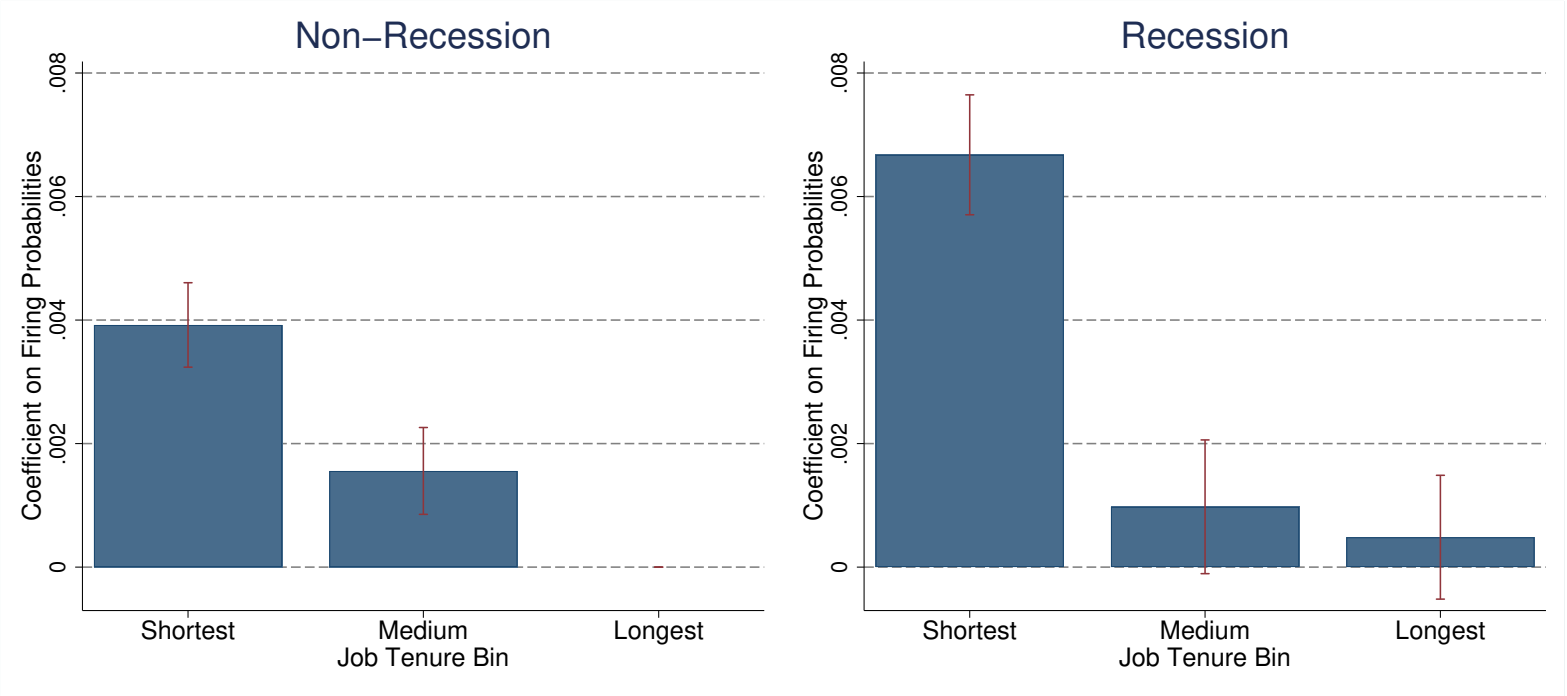
Firing Rate across Wage Bin



Source: IPUMS-CPS, Annual Social and Economic Supplement

- In a recession, overall increase in firing probabilities but the difference across wage groups becomes smaller

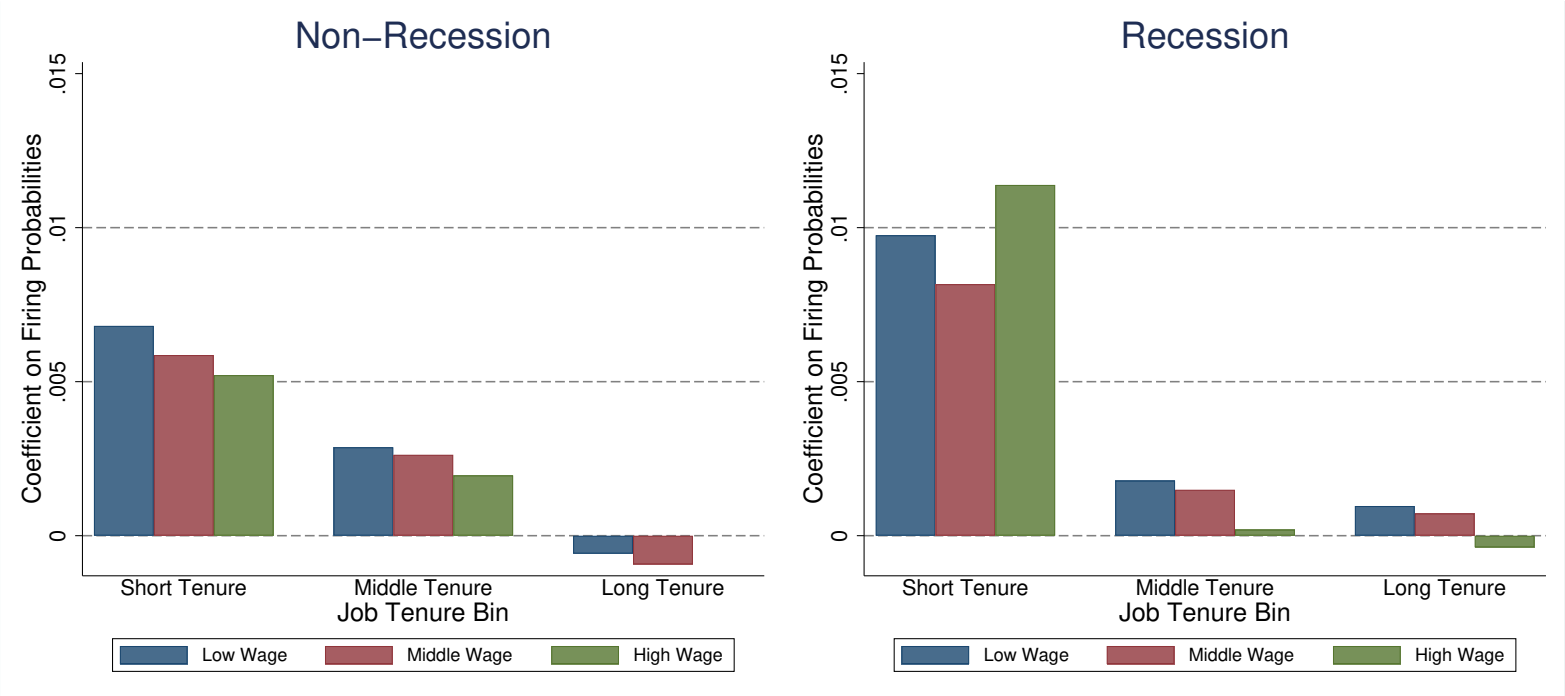
Firing Rate across Job Tenure



Source: IPUMS-CPS, Job Tenure and Occupational Mobility Supplement

- In a recession, only workers with the shortest tenure experience higher firing probabilities
- This is robust to controlling for 1) year trend, 2) metropolitan status, 3) full-part time status, 4) including job leavers

Joint analysis between Tenure and Wage



Source: IPUMS-CPS, ASEC and JTOM Supplement

- Workers with **shortest tenure and highest wage** have the greatest increase in separation probabilities

Data and Definition

- Data: NLSY 79
  - Tracks of weekly work history and weekly working hours
- Dependent variable:
  - 1 if employed in week t and unemployed in week $t+1$ to $t+4$ at all times
  - 0 if employed in week t and employed in any week from $t+1$ to $t+4$
  - Excluded if unemployment is not distinguishable from non-participation
- Key variables:
  - Separation Experience:
    - 1 if the dependent variable becomes 1 anytime in the past
    - 0 otherwise
  - Hourly Wage:
    - Labor income (last year) divided by total hours (last year)
- Controls:
  - Age, Age squared, Race, Occupation, Industry, Gender, Education

Summary Statistics

Estimation Results

Dependent Variable	Separation Probability (in %)		
	(1)	(2)	(3)
Separation Experience	0.246***	0.171***	0.245***
	(0.007)	(0.004)	(0.007)
Recession	0.047***	0.039***	0.046***
	(0.009)	(0.006)	(0.009)
Recession ×	0.264***	0.004	-0.024
Separation Experience	(0.026)	(0.013)	(0.053)
Hourly Wage			0.005***
			(0.000)
Hourly Wage × Recession ×			0.025***
Separation Experience			(0.004)
N	2640113	4902911	2231192
R-squared	0.001	0.001	0.001

(1) and (3) = NLSY Young($25 \leq$ age < 35) samples
(2) = NLSY Entire($25 \leq$ age < 60) samples

- Separation probability tends to be higher for workers with separation experience, especially by more in a recession

Take-away from Empirical Results

- Empirical results from the CPS suggest workers with high wage and short tenures separate from their jobs more frequently during recessions compared to other worker groups
- NLSY results suggest the existence of persistent worker type affecting separation probabilities
- To account for these facts, I need a worker component reducing (increasing) match surplus (separation) but increasing wage *ex ante*
- In the following DMP model, workers are heterogeneous in disutility from work
  - Higher disutility from work
    $\Rightarrow$ Higher wage from Nash-bargaining
    $\Rightarrow$ Lower match surplus
- No cyclical changes in job-finding probabilities. Job-finding prob. regression

Literature Review

- Worker heterogeneity and flows in the labor market
  - Hall and Kudlyak (2019), Ahn and Hamilton (2019), Hall and Schulhofer-Wohl (2018), Kudlyak and Lange (2018)
- Worker type and selective separation
  - Mueller (2018), Morchio (2017), Schmillen and Möller (2012), Michaud (2015), Nakamura (2008), Gibbons and Katz (1991)
- Informational friction in the hiring process and duration dependence
  - Jarosch and Pilossoph (2018), Fernanez-Blanco and Preugschat (2017), Eriksson and Rooth (2014), Faber, Silverman, and von Wachter (2016), Doppelt (2016), Laing (1994), Lockwood (1991)
- Persistent impacts of unemployment shocks on labor market outcomes
  - Yagan (2018), Jarosch (2015), Davis and von Wachter (2011), Jacobson, Louis, LaLonde, and Sullivan (1993), von Wachter, Song, and Manchester (2009)

Cyclical Implications

Model Setting

- Built on standard DMP model with match-specific productivity (ξ) and worker heterogeneity (θ)
- Unit mass of risk neutral workers
  - Heterogeneous type: $\theta \sim m(\theta)$, discrete distribution
  - Low type has low productivity *and* high disutility of work
- Incumbent firm-worker matches use capital as a variable input
- Overhead cost shocks hit the match with Poisson rate Ψ
  - The match has to pay the cost to continue the production
  - If the realized cost is larger than the match surplus, the match dissolves endogenously

Value Functions

- The match surplus is given by

$$\frac{d}{dt}S_t(\theta,\xi) = (r_t + \chi + \delta_t(\theta,\xi))S_t(\theta,\xi) - \pi_t(\theta,\xi) - b(\theta) - \Psi_t\mathbb{C}(S_t(\theta,\xi))$$
$$- p(\tau_t)\eta(1-\eta)\int S_t(\theta,\tilde{\xi})dH(\tilde{\xi})$$

- The wage is given by

$$w_t(\theta,\xi) = (1-\eta)\pi(\theta,\xi) + \eta b(\theta) + \eta(1-\eta)p(\tau_t)\int S_t(\theta,\xi)dH(\tilde{\xi})$$

- Nash bargaining between the firm and the worker over wage
  - At a given wage, low type has lower ξ but higher $b(\theta)$; high type has higher ξ and lower $b(\theta)$
  - Low type has low match surplus and high separation probabilities

Full Characterization

Solution Method

- Use fixed point algorithm to get the steady state without any shock.
- Following Reiter (2010) and Ahn et al. (2018), linearize the HJB equation and the LoM for distribution function at the discrete points of ξ, around the stationary equilibrium
- Apply Blanchard-Kahn condition to pin down forward looking variables (value function and consumption).
- The process for Ψ_t, the frequency of overhead cost shock, is given by the Ornstein–Uhlenbeck process.

$$d(\Psi_t - \Psi) = -\beta(\Psi_t - \Psi)dt + \sigma dW_t$$

where W_t is the standard Brownian motion.

Calibrated Parameters

- Two types of workers,
  $c \sim LogNormal(\mu_c, \sigma_c^2)$, and $\xi \sim LogNormal(-\zeta^2/2, \zeta^2)$

Parameter	Meaning	Value
r^C	Discount rate	0.004
δ	Depreciation rate	0.008
α	DRS parameter with respect to capital input	0.3
$p_U(\tau_U)$	UE rate for unemployed workers	0.45
χ	Exogenous EU rate	0.02
$m(\theta_{high})$	Mass of high surplus type	0.8
$b(\theta_{high})$	Disutility of work for high surplus type	0
μ_c	Mean of log-normal distribution for cost shock	0.23
Mean$[\theta]$	Mean type	1
Mean$[\xi]$	Mean productivity	1

Table: Exogeneously Set Parameters

Calibrated Parameters - 2

- Calibrate key parameters to match steady state moments

Parameter	Meaning	Value
Ψ	Steady state frequency of the overhead cost shock	0.59
σ_c	Standard deviation of log overhead cost	0.64
$b(\theta_{low})$	Disutility of work for low surplus type	1.26
ζ	Standard deviation of log match-specific productivity	0.07
η	Firm share of match surplus	0.71
θ_{high}	Productivity of high surplus type	1.01

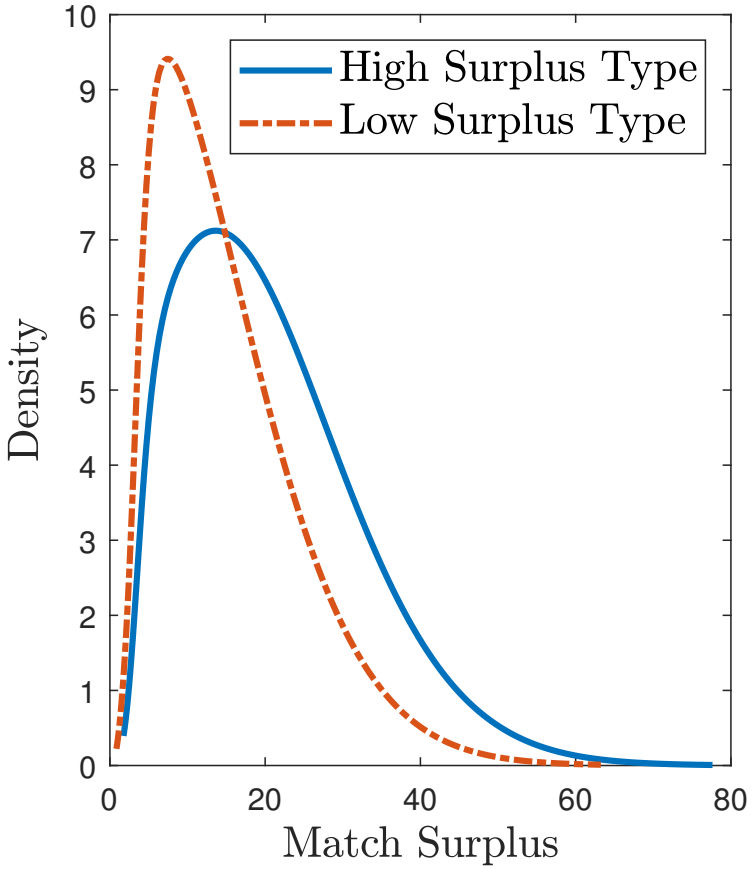
Table: Calibrated Parameters: Steady State

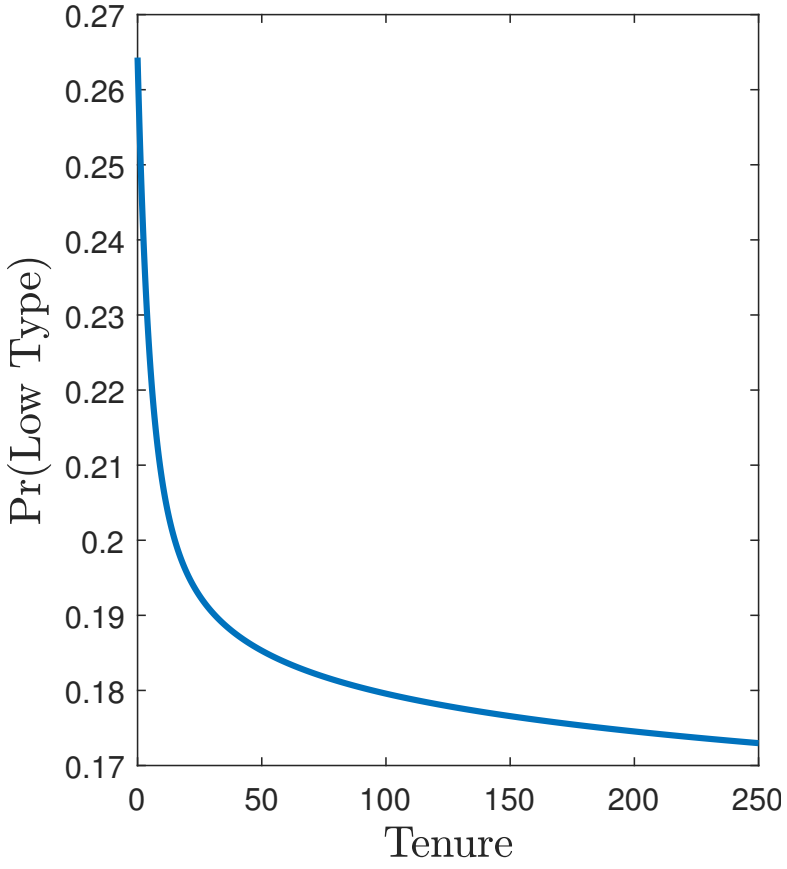
Steady State Moments

Moment	Value	Target
EU rate	0.024	0.025
EU difference between wage groups(3/1)	0.002	0.002
EU difference between wage groups(2/1)	0.001	0.001
EU difference between tenure groups(3/1)	0.008	0.008
EU difference between tenure groups(2/1)	0.005	0.006
Labor share	0.60	0.60
P90/P10 in wage distribution	1.10	2.80
P50/P10 in wage distribution	1.04	1.7

Table: Model and Data Moments: Steady State

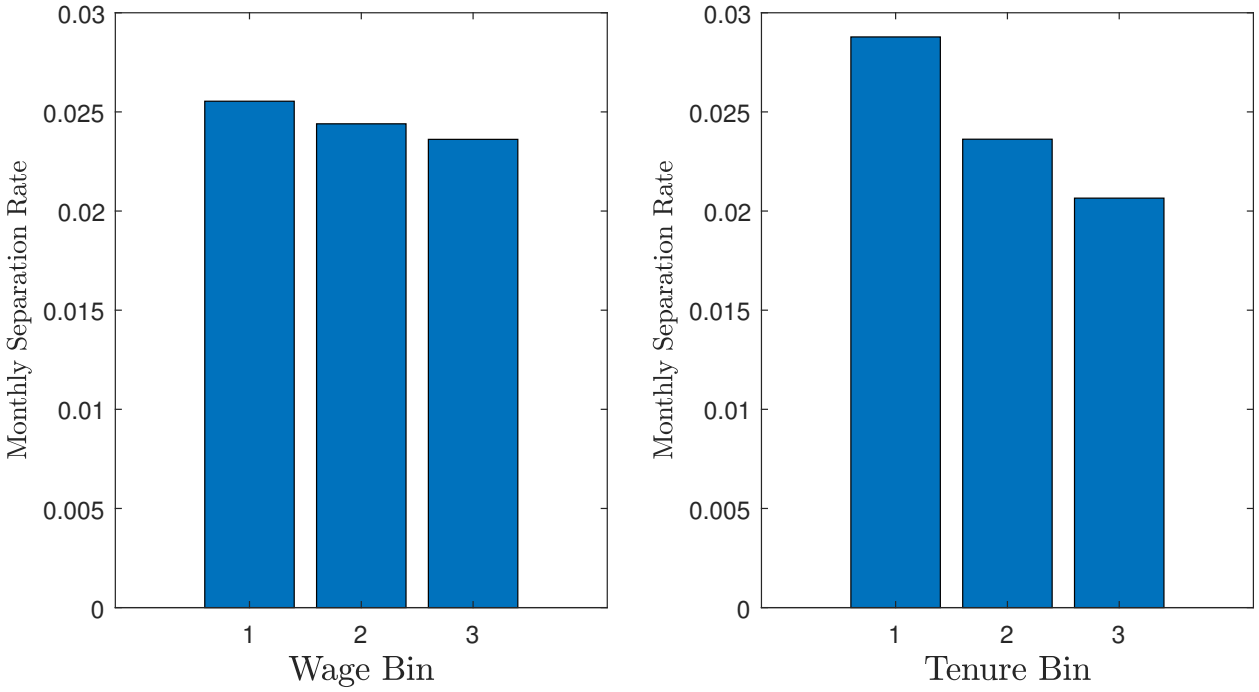
Steady State Distribution





- Because of higher outside value, low surplus type has higher density at lower parts of match surplus distribution
- Especially among short-tenured workers, the fraction of low type decreases rapidly in tenure

Steady State Separation Rates over Wage/Tenure Bin



- Tenure is a stronger predictor of EU transition rates than wage

Increase in Separation across Wage/Tenure Bin

- Shock: 1 std increase in frequency of overhead cost shock
- Largest increase in separation rates for highest wage and shortest tenure bin

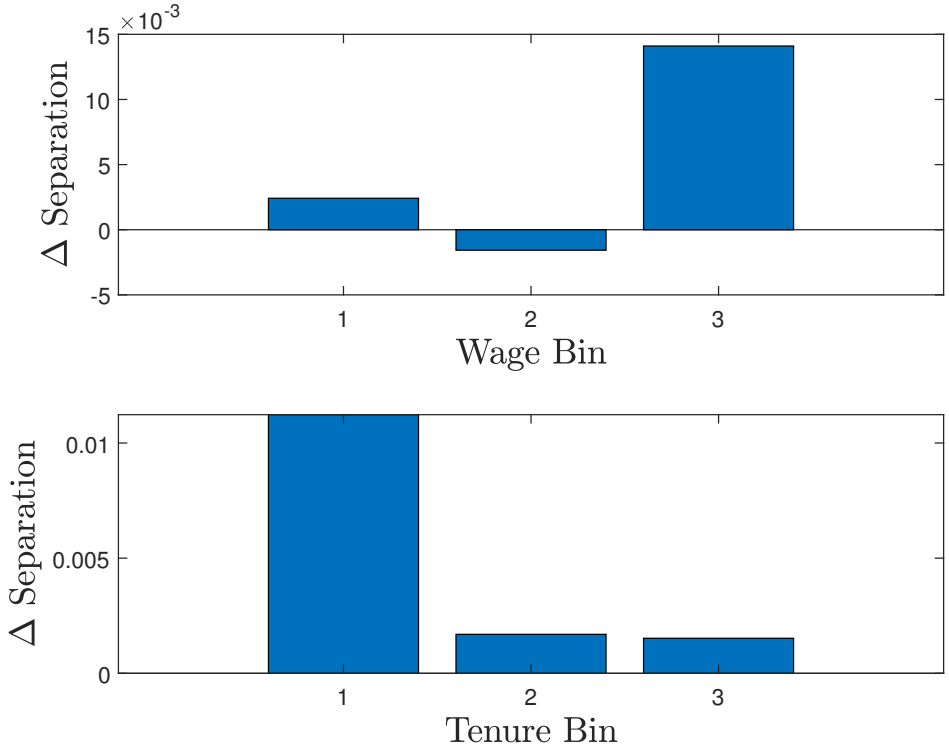


Figure: Relative Increase in Separation Rates over Wage Bin

Labor Market Sorting over the Business Cycle

- Modest composition changes last for several months

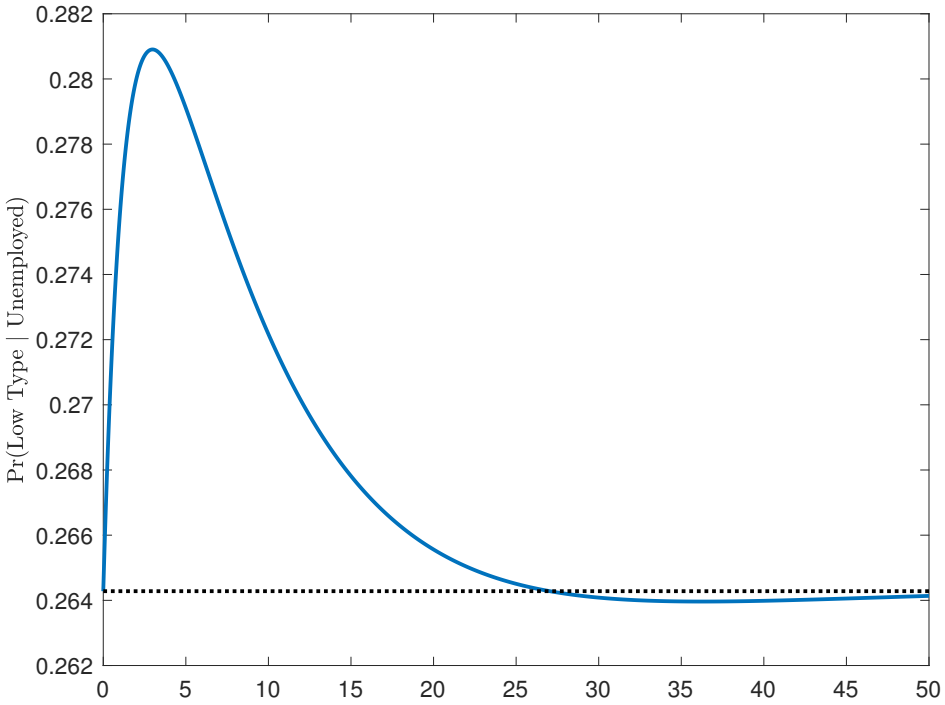
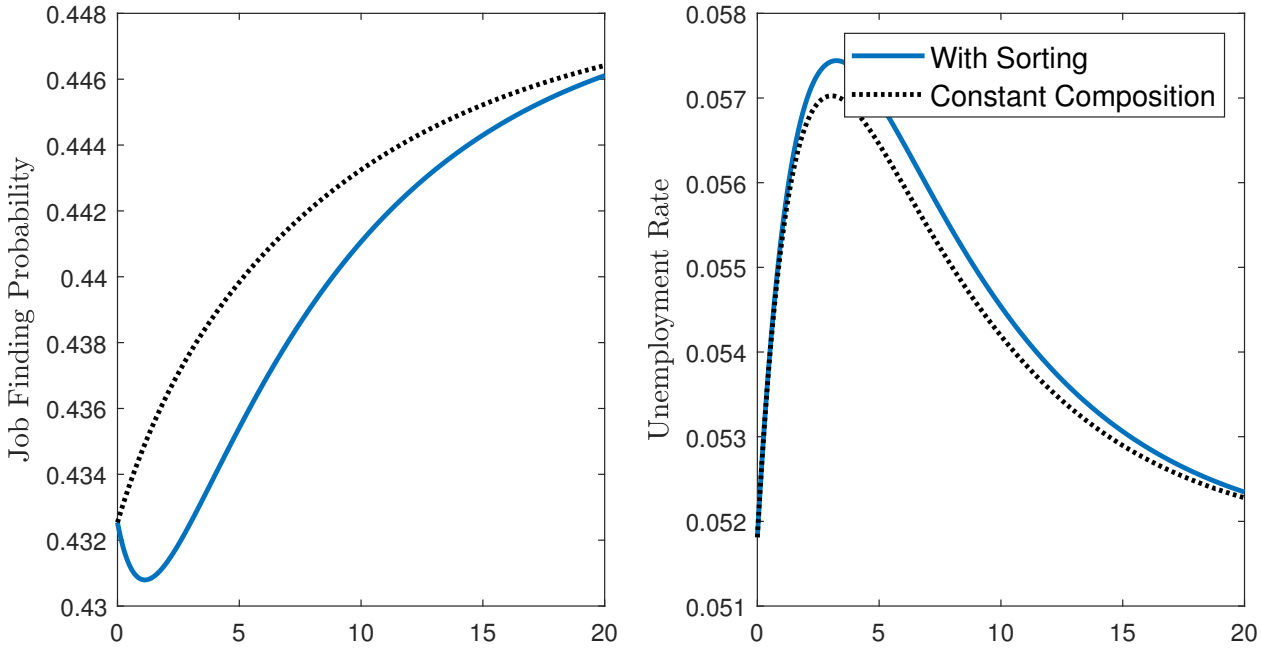


Figure: Fraction of Unemployed Workers that are Low Surplus Type

Without Sorting, Smaller Fluctuations

- Sorting results in hump-shaped responses in job finding rates
- With sorting, 3-month autocorr(UE) $= 0.8759$, std(UE) $= 0.0104$; without it, $= 0.8722$, $= 0.0092$
- As a result, fluctuations in unemployment rates are modestly larger with sorting



Concluding Remarks

- I interpret the separation as a selective process for firms to sort low type workers into unemployment
- CPS and NLSY data exercises imply that workers with low expected match surplus flow into unemployment more frequently in recession periods
- If search is random and new vacancies are more likely to meet unemployed workers, the composition change from sorting decreases expected surplus from creating new match and reduces vacancy creation
  - However, the effect is quantitatively small, because of fast turnover in the US labor market
- The mechanism in this paper can also explain why unemployment shocks lower workers' labor income persistently, and even more in recessions (Yagan, 2019, Gathmann et al., 2017) Empirical Specification

Further Analysis

- Planning to extend the model with
  - on-the-job search in the simplest form
  - ! Separated free entry condition?
  - Discount rate shock as opposed to productivity shock or overhead cost shock
  - Linearized solution vs. Deterministic path after MIT shock
- Planning to include further modeling outcomes
  - Model-generated labor income profile before and after separation event
  - Steady-state vs. in recession

Appendix

Comparison of Separation Measures

	CPS(Shimer)	NLSY79	CPS-ACS	CPS-JTOM
Mean in Normal Times	1.913	0.417	0.622	0.759
Mean in Recession	2.228	0.597	0.801	0.758

Table: Mean of Different Separation Measures (in %)

- Separation Measures from CPS and NLSY are smaller because
  - I exclude separations less than 4 weeks for NLSY and job-leavers for CPS
  - Separation rate measured in Shimer's method tend to be larger than the actual UE transition, because of the non-participation margin

Comparison of Separation Rate over Wage and tenure Groups

Wage Group	Overall	Lowest	Middle	Highest
Mean in Normal Times	0.642	0.764	0.618	0.541
Mean in Recession	0.687	0.780	0.663	0.615

Tenure Group	Overall	Shortest	Middle	Longest
Mean in Normal Times	0.729	1.235	0.620	0.383
Mean in Recession	0.934	1.352	0.565	0.339

- The cross-group mean comparison is consistent with the regression exercise

Go back

Comparison of Separation Rate across Separation Experience

	NLSY79		
	Overall	Never Sep.	Sep. Before
Mean in Normal Times	0.417	0.234	1.040
Mean in Recession	0.597	0.367	1.436
	NLSY79 Young		
	Overall	Never Sep.	Sep. Before
Mean in Normal Times	0.536	0.345	1.601
Mean in Recession	0.940	0.685	3.156

- The cross-group mean comparison is consistent with the regression exercise

Go back

Regression Results for Job-Finding Probabilities: CPS - Wage

Dependent Variable	Separation	Job-Finding Probability (in %)	
	Probabiltiy (in %)	Within 1 Month	Within 2 Months
Non-Recession			
Low Wage	0.20***	-2.22	-1.33
	(0.02)	(1.75)	(1.20)
Middle Wage	0.05**	-2.23	-1.33
	(0.02)	(1.81)	(1.20)
Recession			
Low Wage	0.32***	-8.27***	0.67
	(0.05)	(2.99)	(1.98)
Middle Wage	0.25***	0.23	-3.29
	(0.05)	(3.13)	(2.16)
High Wage	0.21***	-5.55*	-1.04
	(0.05)	(3.27)	(2.18)
N	788,961	5,084	3,643

Regression Results for Job-Finding Probabilities: CPS - Tenure

Dependent Variable	Separation	Job-Finding Probability (in %)	
	Probabiltiy (in %)	Within 1 Month	Within 2 Months
Non-Recession			
Low Tenure	0.53***	0.79	-4.70
	(0.05)	(0.72)	(3.02)
Middle Tenure	0.21***	-0.07	-4.70
	(0.05)	(0.79)	(3.02)
Recession			
Short Tenure	0.90***	-0.86	-1.79
	(0.06)	(0.81)	(3.39)
Middle Tenure	0.16**	-0.91	-8.06
	(0.07)	(1.17)	(4.89)
Long Tenure	0.08	-0.59	-6.25
	(0.07)	(1.17)	(4.88)
N	270,670	2,043	1,934

Regression Results for Job-Finding Probabilities: NLSY79

Dependent Variable	Separation Probability (in %)		Job-Finding Probability (in %)	
	(1)	(2)	(3)	(4)
Separation Experience	0.22***	0.22***	-1.81***	-1.81***
	(0.01)	(0.01)	(0.26)	(0.26)
Hourly Wage	0.00***	0.00***	-0.25***	-0.25***
	(0.00)	(0.00)	(0.02)	(0.02)
Recession	0.07***	0.07***	-0.05	-0.05
	(0.01)	(0.01)	(0.41)	(0.41)
Recession ×	0.25***	-0.09	-0.69	-2.16
Separation Experience	(0.03)	(0.07)	(0.75)	(1.34)
Hourly Wage × Recession ×		0.04***		0.14
Separation Experience		(0.01)		(0.11)
N	2,229,259	2,229,259	62,461	62,461

Go back

Full Characterization of the Economy

$$
\begin{aligned}
\frac{d}{dt} g_t(\theta,\xi) &= p(\tau_t) u_t(\theta) h(\xi) - (\chi + \delta_t(\theta,\xi)) g_t(\theta,\xi) \\
\frac{d}{dt} K_t + u'^{-1}(m u_t) + c_v v_t &= \left(\frac{\alpha}{r_t}\right)^{\frac{\alpha}{1-\alpha}} \sum_\theta \int (\xi\theta)^{\frac{1}{1-\alpha}} g_t(\theta,\xi) d\xi \\
&\quad - \sum_\theta \Psi_t \int \mathbb{C}(S_t(\theta,\xi)) g_t(\xi,\theta) d\xi - \delta K_t \\
K_t &= \left(\frac{\alpha}{r_t}\right)^{\frac{1}{1-\alpha}} \sum_\theta \int (\xi\theta)^{\frac{1}{1-\alpha}} g_t(\theta,\xi) d\xi \\
d(\Psi_t - \Psi) &= -\beta(\Psi_t - \Psi) dt + \sigma dW_t \\
\frac{d}{dt} S_t(\theta,\xi) &= (r_t + \chi + \delta_t(\theta,\xi)) S_t(\theta,\xi) - \pi_t(\theta,\xi) - b(\theta) - \Psi_t \mathbb{C}(S_t(\theta,\xi)) \\
&\quad - p(\tau_t)\eta(1-\eta) \int S_t(\theta,\tilde{\xi}) dH(\tilde{\xi}) \\
\delta_t(\theta,\xi) &= \Psi_t (1 - F(S_t(\theta,\xi))) \\
\frac{d}{dt} m u_t &= (\rho + \delta - r_t) m u_t \\
c &= q(\tau_t) \int \sum_\theta \frac{u_t(\theta)}{u_t} \eta S_t(\theta,\xi|\Omega) dH(\xi)
\end{aligned}
$$

Go back

Job Finding and Job Separation Measure

- LEHD - J2J
  - NE Persist Hire/ EN Persist Separation in MSA level (Employment = UI-covered jobs)
  - NE Persist Hire in t of individual i at firm j if
    1. i does not have any income source in quarter $t-1$
    2. Income from j in quarter t and $t+1$ appears to be the largest among all income sources of i
  - EN Persist Separation in t of individual i at firm j if
    1. Income from firm j in quarter t and $t-1$ appears to be the largest among all income sources of i
    2. i does not have any income source in quarter $t+1$
- Use MSA variation in Non-J2J hire/separation to construct the following measures over MSA m

$$s_{m07-10} = \frac{\#\{\text{EN Persist Separations in m from 07 to 10}\}}{\#\{\text{Employed Workers in m in 06}\}}$$

$$f_{m07-10} = \frac{\#\{\text{NE Persist Hires in m from 07 to 10}\}}{\#\{\text{Unemployed Workers in m in 06}\}}$$

Estimation Specification

- In 2007-2010, the regression specification is

$$y_m = \beta_0 + \beta_1 hpg_m + \gamma Bartik_m + \varepsilon_m$$

  where $Bartik_m$ is Bartik control for local labor market; local labor demand predicted by industrial composition at the beginning of the period, and y_m is the MSA-level EN Persist separation and NE Persist hire rate measures

- In other periods, the regression specification is

$$y_m = \beta_0 + \beta_1 hpg_m + \beta_2 hpg_{m,07-10} + \gamma Bartik_m + \varepsilon_m$$

  Note that I additionally control for contemporaneous housing price shocks because of serial correlation

- Each sample is weighted by the employment/unemployment for separation rate/finding rate, respectively

Estimation Results

Dependent Variable	OLS			IV: Saiz instrument		
	2004-06	2007-10	2011-14	2004-06	2007-10	2011-14
NE Persist Hire Rate	5.183***	2.312	8.062***	12.856	-8.363	5.787***
	(1.575)	(3.676)	(0.584)	(13.260)	(11.865)	(1.841)
EN Persist Separation Rate	-0.005	-0.101**	-0.091**	0.739*	-0.246**	-0.122
	(0.062)	(0.045)	(0.038)	(0.431)	(0.120)	(0.115)

- When the housing price shock is instrumented with Saiz elasticity of housing supply, housing net worth shock has persistently negative impacts on job finding in 2011-2014
- The effect is sizeable; ten percent decrease in housing prices during the GR is associated with an increase in monthly EN Persist separation rate by 0.008 in the same period and a decrease in monthly NE Persist hire rate by 0.15 from 2011 to 2014

Go back